

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1}{}^\dagger \begin{array}{|c|c|} \hline -\frac{3\Upsilon_1 k^2}{8} & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1}{}^{\alpha\beta\chi} \begin{array}{|c|c|} \hline 0 & 3k^2 \overset{(4)}{\kappa}_{15} \\ \hline \end{array} \quad \mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{K}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} \begin{array}{|c|c|c|c|} \hline -\frac{\Upsilon_2 k^2}{8} & \frac{\Upsilon_2 k^2}{4\sqrt{2}} & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} \begin{array}{|c|c|c|c|} \hline \frac{\Upsilon_2 k^2}{4\sqrt{2}} & -\frac{\Upsilon_2 k^2}{4} & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha} \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha} \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} \begin{array}{|c|c|} \hline \frac{9\Upsilon_2 k^2}{8} & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi} \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}$$

$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1}{}^\dagger \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(0^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1}{}^{\alpha\beta\chi} \begin{array}{|c|c|} \hline 0 & \frac{1}{3k^2 \overset{(4)}{\kappa}_{15}} \\ \hline \end{array} \quad \mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} \begin{array}{|c|c|c|c|} \hline \frac{1}{3\text{Det}(1^+)} & -\frac{\sqrt{2}}{3\text{Det}(1^+)} & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} \begin{array}{|c|c|c|c|} \hline -\frac{\sqrt{2}}{3\text{Det}(1^+)} & \frac{2}{3\text{Det}(1^+)} & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha} \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha} \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(2^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi} \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
\Upsilon_1 &== 4 \overset{(4)}{\kappa}_1 - 6 \overset{(4)}{\kappa}_{15} + 3 \overset{(4)}{\kappa}_3 \text{ \& } \Upsilon_2 == 2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 \text{ \& } \text{Det}(0^+) == -\frac{3}{8} k^2 (4 \overset{(4)}{\kappa}_1 - 6 \overset{(4)}{\kappa}_{15} + 3 \overset{(4)}{\kappa}_3) \text{ \& } \\
\text{Det}(1^+) &== \frac{3}{8} k^2 (-2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_3) \text{ \& } \text{Det}(2^+) == \frac{9}{8} k^2 (2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3)
\end{aligned}$$

Lagrangian

$$\begin{aligned}
&\overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta} \partial^{\chi} \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\alpha\chi} - 3 \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} - \\
&\frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} - \frac{9}{2} \overset{(4)}{\kappa}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\chi\alpha} + \frac{5}{4} \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\chi\alpha} - \\
&\frac{3}{2} \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{4} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{1^+}^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$
$\mathcal{J}_{1^+}^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$
$2 \mathcal{J}_{1^+}^{\#1\alpha\beta} + \mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\chi \mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$	5	$ \begin{aligned} &3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \\ &4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} == \\ &3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \\ &2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta} \end{aligned} $
Total # constraint(s):	14	
Resolved unitarity condition(s):	True	